

Joel Bhattarai

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EDUCATION

University of Michigan

Bachelor of Science in Engineering in Mechanical Engineering

Ann Arbor, MI

May 2026

EXPERIENCE

Principal Investigator

NASA L'SPACE Program - Proposal Writing and Evaluation Experience

May 2025 – September 2025

- Independently generated a new technology within the areas of Autonomy, Robotics, and AI that was selected for a full proposal by a NASA panel
- Led a group of ten interdisciplinary students to a full proposal and an NTR (New Technology Report)
- Facilitated advancement of proposed technology from concept to final proposal
- Served as the primary reviewer on a NASA proposal review board

Lead Systems Engineer

NASA L'SPACE Program - Mission Concept Academy

May 2024 – September 2024

- Managed a team of engineering students to a successful PDR
- Designed a rover to accomplish practical missions on the surface of the moon
- Completed skills modules on Heat Transfer, Siemens NX, Teaming, Engineering Requirements, Project Management, Systems Topics, Risk Management

Thermal Interface/Battery Pack Design Team Member

Dearborn, MI

Formula SAE - Electric

September 2023 – January 2024

- Collaborated with engineers in a team to develop an electric race car
- Designed cooling solutions for the battery pack, electric motors, etc.
- Designed the battery pack for optimal cooling

PROJECTS

Inverted Pendulum Stabilization

November 2025

Automotive Laboratory

University of Michigan

- Modeled physical behavior of inverted pendulum setup in state space for control design.
- Numerically determined multiple gain values using pole placement for a full-state feedback controller.
- Leveraged Simulink to apply solution to physical hardware and eliminated settling time based on real-world performance.

SKILLS

Software & Simulation: SolidWorks, Siemens NX, HyperMesh, HyperWorks, OptiStruct, ANSYS, LaTeX

Controls & Computing: MATLAB, Simulink, LabVIEW, Arduino, C/C++

Manufacturing & Lab: Machining (Mill & Lathe), Metalworking, Waterjetting, Geometric Dimensioning & Tolerancing (GD&T)

Testing & Systems: Oscilloscopes, Multimeters, Strain Gauges, Thermocouples, Systems Engineering

COURSES

Relevant Coursework

- | | | |
|----------------------------------|------------------------------|---------------------------|
| • Solid Mechanics and Statics | • Fluid Mechanics | • Automatic Control |
| • Thermodynamics | • Heat Transfer | • Energy Infrastructure |
| • Dynamics and Vibrations | • Control of Dynamic Systems | • Modern Energy Materials |
| • Design and Manufacturing I-III | • Behavior of Materials | |
| • Finite Elements | • Electric Circuits | |